KENTARO ISHII

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Current status

Center for the Neurobiology of Addiction, Pain, and Emotion, Department of Anesthesiology and Pain Medicine, Department of Pharmacology, University of Washington
in the laboratory of Prof. Garret Stuber 2019-present

Education

Ph. D Department of Applied Biological Chemistry, Graduate School of Agricultural and Life Sciences, 2019, The University of Tokyo
M.S. Department of Applied Biological Chemistry, Graduate School of Agricultural and Life Sciences, 2016, The University of Tokyo
B.S. Department of Applied Biological Chemistry and Technology, 2014, The University of Tokyo

Experience

The University of Washington, Center for the Neurobiology of Addiction, Pain, and Emotion, Department of Anesthesiology and Pain Medicine, Department of Pharmacology

Acting instructor in the laboratory of Prof. Garret Stuber 2024-
- Characterized morphine responsive neurons in the mouse brain
- Constructing pre-processing pipeline for whole brain imaging

Research scholar in the laboratory of Prof. Garret Stuber 2019-2024
- Characterized estradiol sensitive neurons in the paraventricular hypothalamus.
- Identified neurons in the medial preoptic area which regulates sexual motivation

The University of Tokyo, Graduate School of Agricultural and Life Sciences, Department of Applied Biological Chemistry
Graduate Student (Doctor-Course) in the laboratory of Prof. Kazushige Touhara 2016-2019
- Identified neural circuit conveying a juvenile pheromone information in mice.
- TA (assisted experiment, led discussion, prepared and deliverer lectures) in hands-on biochemistry experiment course.

The University of Tokyo, Graduate School of Agricultural and Life Sciences, Department of Applied Biological Chemistry
Graduate Student (Master-Course) in the laboratory of Prof. Kazushige Touhara 2014-2016
- Identified neural circuit conveying sex pheromone information in mice.
- Assisted in one under graduate-level training course.

The University of Tokyo, Department of Applied Biological Chemistry and Technology
Under Graduate Student in the laboratory of Prof. Kazushige Touhara 2013-2014
- Identified the type of hypothalamic neuron responding to sex pheromone stimuli in mice.

Awards

Dean award of the Department of Applied Biological Chemistry (Doctor-course) 2019
Young Investigator Travel Award of the 17th International Symposium on Olfaction and Taste 2016
Poster award of 16 th Univesity of Tokyo Biology Symposium 2016
Dean award of the Department of Applied Biological Chemistry (Master-course) 2016

Research Funding and Fellowship

Overseas Post-Doctoral Fellowship from Uehara Memorial Foundation
4,100,000 JPY per year 2021-2022
Overseas Research Fellowship from JSPS 55,700 USD per year 2019-2021
Research Fellowship for Young Scientists from JSPS 2,400,000 JPY per year 2016-2019
Early-Career Collaborative Research Project from Kobe University 2,000,000 JPY per year 2025-2026

Pending

NIH/NIDA R03 (START Program, PAS-27-028)
Developmental Emergence of Opioid-Responsive Neuronal Ensembles
Role: PI $200,000 direct costs (2 yrs) 2027-2029
NIH R21 (PA-25-304)
Whole-Brain Neural Ensemble Mapping of Kratom Alkaloids
Role: PI $275,000 direct costs (2 yrs) 2027-2029

Presentations and Publications

Selected Publications

1. **Ishii, K.K.***, Osakada, T.*, Mori, H., Miyasaka, N., Yoshihara, Y., Miyamichi, K., and Touhara, K. (2017). A labeled-line neural circuit for pheromone-mediated sexual behaviors in mice. **Neuron 95**, 123–137.e8.
2. Osakada, T.*, **Ishii, K.K.***, Mori, H.*, Eguchi, R., Ferrero, D.M., Yoshihara, Y., Liberles, S.D., Miyamichi, K., and Touhara, K. (2018).
   Sexual rejection via a vomeronasal receptor-triggered limbic circuit. **Nat. Commun**. **9**, 4463.
3. Itakura, T., Murata, K., Miyamichi, K., **Ishii, K.K.**, Yoshihara, Y., and Touhara, K. (2022).
   A single vomeronasal receptor promotes intermale aggression through dedicated hypothalamic neurons. **Neuron 110**, 2455–2469.e8.
4. **Ishii, K.K.**, Hashikawa, K., Chea, J., Yin, S., Fox, R.E., Kan, S., Shah, M., Zhou, Z.C., Navarrete, J., Murry, A.D., Szelenyi, E.R., Golden, S.A., and Stuber, G.D. (2025).
   Post-ejaculatory inhibition of female sexual drive via heterogeneous neuronal ensembles in the medial preoptic area. **eLife 12**, RP91765.
5. Simon, R.C., Fleming, W.T., Briones, B.A., Trzeciak, M., Senthilkumar, P., **Ishii, K.K.**, Hjort, M.M., Martin, M.M., Hashikawa, K., Sanders, A.D., Golden, S.A., and Stuber, G.D. (2025).
   Opioid-driven disruption of the septum reveals a role for neurotensin-expressing neurons in withdrawal. **Neuron 113**, 2325–2343.e9.
6. Hashikawa, K., Hashikawa, Y., Briones, B., **Ishii, K.**, Liu, Y., Rossi, M.A., Basiri, M.L., Chen, J.Y., Ahmad, O.R., Mukundan, R.V., Johnston, N.L., Simon, R.C., Soetedjo, J.C., Siputro, J.R., McHenry, J.A., Palmiter, R.D., Rubinow, D.R., Zweifel, L.S., and Stuber, G.D. (2026).
   Esr1-dependent signaling and transcriptional maturation in the medial preoptic area of the hypothalamus shapes the development of mating behavior during adolescence. **eLife 14**, RP106347.

Preprints

9. **Ishii, K.K., Zhang, Y., Ren, L., de Lecea, A., Bakshi, S., Haygood, A., Chujo, S., Stevenson, H.P., Schwartz, E., Zhou, Z.C., Szelenyi, E.R., Golden, S.A., Kessler, D., and Stuber, G.D. (2025).**
   Hierarchical and spatial mapping of whole-brain c-Fos activity reveals distinct opioid and withdrawal neuronal ensembles. **bioRxiv** 2025.09.21.677198. *accepted in* ***Neuron***

Selected Presentation

1. “An allocated neural circuit in the amygdala-hypothalamus axis for sex pheromone-mediated behavior in female mice”

Kentaro Ishii, Kazunari Miyamichi, Kazushige Touhara. **"**Optogenetic Approaches to Understanding Neural Circuits & Behavior" Gordon Research Conference 2016**.** Poster presentation.

2. "Neural circuitry for positive and negative modulation of pheromone-mediated female sexual behavior"
Kentaro Ishii, Takuya Osakada, Hiromi Mori, Kazunari Miyamichi, Kazushige Touhara. The seventh Cold Spring Harbor conference on Neuronal Circuits 2018. Poster presentation
3. "Dissecting sensory and motivational signals in the MPOA during mating in female mice"
Kentaro K Ishii, Adam Gordon-Fennell, Koichi Hashikawa, Jane Chea, Shihan Yin, Rebecca Erin Fox, Garret Stuber. "Hypothalamus" Gordon Research Conference 2022. Poster presentation.
4. "Brain wide neuronal ensembles engaged by opioid"
Kentaro K Ishii, Garret Stuber.
Invited guest at Kobe University. 11/18/2024. Oral presentation.
5. "Inhibitory control of sexual motivation by the medial preoptic area"
Kentaro K Ishii, Garret Stuber.
Invited guest at Tokyo University. 11/20/2024. Oral presentation.
6. "Whole brain analysis of opioid responding ensemble using improved multiple comparison correction algorithm "
Kentaro K Ishii, Garret Stuber.
Invited guest at Juntendo University. 11/21/2024. Oral presentation.
7. "Brain-wide mapping of neural ensembles during morphine addiction using a novel error rate and testing strategy"
Kentaro K Ishii, Daniel Kessler, Garret D. Stuber
"2025 CNCM conference The Changing Brain", UC Irvine. 8/18/2025. Poster presentation.
8. "Systematic investigation of opioid-responsive neurons in the PVT" (Chair: Dr. Lee-Yuan Liu-Chen),
Kentaro K Ishii, Daniel Kessler, Garret D. Stuber
International Narcotics Research Conference 2026, Denver, CO. 7/16/2026. Oral presentation.
9. "Mapping whole-brain opioid ensembles with light sheet microscopy"
Kentaro K Ishii
LIFECANVAS PATHWAYS, LifeCanvas webinar series. 11/5/2026.

Reviews and Commentaries

1. "The neural circuit determining the valence of odors"
Kentaro Ishii, Kazushige Touhara *Journal of Japan Association on Odor and Environment* Vol. 45, No.4, 2015 (in Japanese)
2. "Neural circuits regulating sexual behaviors via the olfactory system in mice."
Ishii, K.K., and Touhara, K. (2019). *Neurosci. Res*.

Book Chapters

1. "Encyclopedia of Living Things, Scents, and Pheromones"
Ishii, K. "The amygdala". In K. Touhara, Y. Niimura, Y. Yoshihara, M. Yokosuka, T. Kikusui, & M. Okamoto (Eds.), *Encyclopedia of animals and odors & pheromones* (Chap. 2.5.5, pp. 118-121). Asakura Publishing.

Lectures

1. "Light sheet data analysis on the UW high-performance computing cluster"
Kentaro Ishii, NAPE computation seminar series. University of Washington. 1/20/2023.
2. "Inhibitory control of female sexual motivation by the medial preoptic area"
Kentaro Ishii, Membrane and Muscle Biophysics Seminar. University of Washington. 3/11/2023.
3. "Light sheet data preprocessing using the high-performance computing cluster"
Kentaro Ishii, LIGHT SHEET IMAGING WORKSHOP. University of Washington. 6/14/2024.
4. "Neural substrate for post-ejaculatory inhibition of female sexual drive"
Kentaro Ishii, Neuroscience program Winter 2025 seminar series. University of Washington. 3/10/2025.

Workshop

1. "IMAGING &GENETICS WORKSHOP 2021"
Charles Zhou, Abbie Schindler, Jane Chen, Kentaro Ishii, Sean Piantadosi, Adam Gordon-Fennell, Raaj Gowrishankar, Madelyn Hjort, Lusine Gomtsian
"Virus injection, GRIN lens, and headcap implantation surgery" University of Washington Center for the Neurobiology of Addiction, Pain, and Emotion. Participated in organizing and launching the workshop. Conducted a lecturer for free-moving *in vivo* imaging demonstration, experiment design crafting seminar speaker and intracranial lens implant surgery demonstration.

2. “IMAGING &GENETICS WORKSHOP 2022”
   Charles Zhou, Abbie Schindler, Jane Chen, Kentaro Ishii, Sean Piantadosi, Adam Gordon-Fennell, Raaj Gowrishankar, Madelyn Hjort, Lusine Gomtsian
   “Virus injection, GRIN lens, and headcap implantation surgery” University of Washington Center for the Neurobiology of Addiction, Pain, and Emotion. Participated in organizing the workshop. Conducted a lecturer for free-moving *in vivo* imaging demonstration, experiment design crafting seminar speaker and intracranial lens implant surgery demonstration.
3. “IMAGING &GENETICS WORKSHOP 2023”
   Charles Zhou, Abbie Schindler, Jane Chen, Kentaro Ishii, Sean Piantadosi, Adam Gordon-Fennell, Raaj Gowrishankar, Madelyn Hjort, Lusine Gomtsian
   “Virus injection, GRIN lens, and headcap implantation surgery” University of Washington Center for the Neurobiology of Addiction, Pain, and Emotion. Participated in organizing the workshop. Conducted a lecturer for free-moving *in vivo* imaging demonstration, experiment design crafting seminar speaker and intracranial lens implant surgery demonstration.
4. “LIGHT SHEET IMAGING WORKSHOP 2024”
   Charles Zhou, Kentaro Ishii, Eric Szelenyi, Jingyi Chen, Jovana Navarrete, Sam Golden, Lusine Gomtsian
   “Demonstration of tissue clearing using the LifeCanvas SmartBatch system and preprocessing pipeline using high performance cluster computer” University of Washington Center for the Neurobiology of Addiction, Pain, and Emotion. Participated in organizing the workshop. Prepared a 2-hour lecture and led a hands-on training course for whole brain imaging data analysis and processing. Produced and shared multiple recorded lectures for data analysis and processing.

References

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